Software Requirements Specification for FPVR: subtitle describing software

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Revision History

Date	Developer(s)	Changes
Oct 16, 2026	Sankalpa	Adapted the SRS document structure to Meyer's template.
Oct 18, 2026	Sankalpa	Completed the Goals (G) section of the SRS.

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(G) Goals

G.1 Context and Overall Objectives

Learning to operate a first-person-view (FPV) drone presents challenges for beginners, particularly when practicing with physical equipment. New pilots must develop coordinated flight-control skills while managing the possibility of crashes, equipment damage, and unsafe flight conditions. These challenges can make initial training costly and discourage inexperienced users from gaining practical experience.

FPVR is a virtual reality (VR)-based FPV drone training simulator designed to provide a safe, accessible, and structured environment for developing foundational drone piloting skills. The system will allow users to control a simulated drone, practice flight manoeuvres, and complete progressively challenging courses without requiring a physical drone.

The training environment will include free-flight practice, multiple obstacle-course levels, distinct virtual environments, checkpoints, and performance evaluation. Users will receive feedback on their flight performance through measurable indicators such as completion time, checkpoint progress, and performance ratings. An optional intelligent feedback component may provide further analysis of user performance.

G.2 Current situation

Learning to operate a first-person-view (FPV) drone requires beginners to develop coordinated control skills while interpreting visual feedback from the drone's perspective. Practicing with physical drones introduces risks of crashes, equipment damage, and additional costs, making simulation a valuable alternative for introductory training.

Several established FPV drone simulators, including Liftoff: FPV Drone Racing [3] and VelociDrone[4], allow users to practice flight manoeuvres, complete racing courses, and explore virtual environments through conventional PC-based interfaces. These platforms provide extensive training and recreational capabilities, with a strong emphasis on realistic flight behaviour, racing, and freestyle flying. However, their traditional desktop experiences do not provide the same interaction environment as a dedicated virtual reality (VR) headset.

VR-based alternatives also exist. Liftoff: XR Drone [5] introduces accessible drone flight using Meta Quest controllers, while DVR Simulator [6] and Drone Blitz [7] offer VR flight challenges and racing or performance-tracking features. These applications demonstrate that immersive drone simulation is already feasible, but they differ in their emphasis on casual exploration, racing, and structured practice.

FPVR aims to build on these existing approaches by focusing specifically on a progressive,

beginner-oriented VR training experience. Rather than treating flight practice and course completion as isolated activities, FPVR intends to connect free-flight familiarization, increasingly challenging obstacle courses, checkpoint-based objectives, and post-flight performance reviews into a continuous learning process. By allowing users to evaluate their results and repeat challenges, the system aims to help beginners recognize their mistakes and track improvements over successive attempts.

The intended contribution of FPVR is therefore not the introduction of drone simulation or VR flight itself, but the integration of immersive flight practice with structured progression and measurable feedback. The effectiveness of this approach will require evaluation with potential users.

G.3 Expected Benefits

Safe and Repeatable Practice (G1): FPVR will allow users to practice drone operation without exposing a real drone or surrounding property to collision-related damage. Users can recover from simulated crashes and repeat exercises without the costs associated with damaging physical equipment.

Progressive Skill Development (G2): Rather than relying exclusively on unrestricted flight, FPVR will offer structured challenges of varying difficulty. Users can freely select levels based on their experience and practice increasingly demanding manoeuvres at their own pace.

Performance Awareness (G3): After completing a training level, users will be able to review their performance through recorded measures such as completion time, checkpoint progress, and collisions. This feedback is intended to help users identify areas for improvement and encourage deliberate practice.

Accessible Introductory Training (G4): FPVR will accommodate users with limited or no prior FPV flying experience by providing a free-flight environment where they can become familiar with flight controls before attempting structured challenges.

These benefits concern the intended training experience. Any claim that FPVR improves actual physical-drone piloting ability will require separate validation and is not assumed by this specification.

G.4 Functionality overview

FPVR will provide the following capabilities to support its goals.

Table 1: FPVR Functionality Overview

Functionality	**Description**	**Initial Scope**
VR Flight Experience	First-person simulated drone flight through a virtual reality headset.	Core
Free-Flight Practice	An open-ended practice environment where users can freely operate the drone without course objectives, timers, or scoring.	Core
Obstacle-Course Levels	Structured flight challenges of varying difficulty. Users may select any available level regardless of previous completion.	Core
Different Environments	Multiple virtual environments for flight practice and obstacle courses. Different levels may share the same environment.	Core
Checkpoint System	Designated course checkpoints used to track progress toward level completion.	Core
Crash Recovery	Following a simulated crash, users can choose to restart the level or respawn at the most recently reached checkpoint.	Core
Timing and Scoring	Records completion time and calculates a score based on defined performance measures during obstacle-course challenges.	Core
Flight Review	Provides a summary of flight performance after completing a level, including completion time, collisions, checkpoint results, and score.	Core
Flight Rating	Provides a separate performance rating based on defined flight measures. The evaluation criteria remain to be determined.	Core
Intelligent Feedback	Potential AI-based analysis of flight performance to provide additional observations and recommendations, subject to feasibility evaluation.	Planned
Controller Input	Supports drone control using Meta Quest controllers or a compatible Xbox controller.	Core

The exact scoring and rating criteria, controller mappings, checkpoint and respawn behaviour, and requirements for AI-based feedback will be specified in the System section.

G.5 High-level usage scenarios

This section describes the primary interactions between users and FPVR during flight practice, obstacle-course completion, crash recovery, and performance evaluation. These scenarios illustrate the intended system behaviour and support the goals defined in Section G.1.

G.5.1 Scenario 1: Initial Free-Flight Practice

A first-time user launches FPVR and selects the free-flight mode using either Meta Quest controllers or a compatible Xbox controller. The user enters a virtual environment where they can practice basic flight manoeuvres, such as ascending, descending, turning, and navigating through the environment.

Unlike structured obstacle-course levels, free-flight mode has no checkpoints, timers, or scoring. Users can explore and practice at their own pace without specific completion objectives.

If the drone experiences a simulated crash, the user can reset its position and continue practicing. The user may leave free-flight mode and return to the level-selection interface when desired.

This scenario supports G1 and G4 by providing a safe, repeatable environment for becoming familiar with FPV drone controls.

G.5.2 Scenario 2: Completing an Obstacle-Course Level

A user selects an obstacle-course level from the available challenges. All levels are accessible regardless of previous completion, allowing users to choose courses appropriate to their experience and preferences.

Each level contains obstacles and a sequence of checkpoints that must be completed in the designated order. The system tracks the user's progression, total elapsed time, and time spent navigating between checkpoints. Upon reaching a checkpoint, the current section timer resets while the total elapsed time continues accumulating.

The user navigates through the checkpoints in sequence until reaching the finish. After successful completion, the system calculates a numerical score and a separate performance rating based on the recorded flight results.

The user can then review their performance, retry the level, or select another challenge.

This scenario supports G2 by providing structured flight challenges of varying difficulty with measurable outcomes.

G.5.3 Scenario 3: Recovering from a Simulated Crash

While attempting an obstacle-course level, the user loses control of the drone and experiences a simulated crash. The system records the collision and provides two recovery options: restart the entire level or respawn at the most recently reached checkpoint.

If the user selects respawn, the drone returns to the last successfully reached checkpoint. If no checkpoints have been reached, the drone returns to the level's starting position.

The current section timer resets upon respawning, allowing the user to attempt the section again. However, the total elapsed time continues accumulating, including the time spent on failed attempts. The recorded collision also negatively affects the final score.

The user can continue attempting the course without restarting the FPVR application.

This scenario supports G1 and G2 by allowing users to recover from mistakes while maintaining meaningful performance consequences.

G.5.4 Scenario 4: Reviewing Flight Performance

After successfully completing an obstacle-course level, the user receives a flight review summarizing their performance.

The review includes the total completion time, checkpoint results, collision count, and numerical score. The total completion time includes time spent on failed sections and subsequent attempts following checkpoint respawns.

The system also presents a separate overall performance rating based on defined evaluation criteria. The user can examine these results to identify areas for improvement before attempting another flight challenge.

This scenario supports G3 by providing measurable feedback intended to encourage deliberate practice and improvement.

The exact scoring formula, rating categories, and penalties associated with collisions must be established in S.2.

G.5.5 Scenario 5: AI-Assisted Flight Feedback (Planned: Subject to Feasibility)

If AI-assisted feedback is included in the final implementation, the system will automatically provide additional performance analysis as part of the post-level flight review.

Using available flight-performance data, the system would identify areas of difficulty and provide understandable observations or recommendations. For example, the feedback might highlight repeated collisions or difficulties navigating particular obstacle sections.

The intelligent feedback component is intended to complement the standard numerical score and performance rating rather than replace them.

The feasibility, data requirements, analysis approach, and evaluation criteria for AI-assisted feedback remain to be determined. The standard flight review will remain available independently of this optional capability.

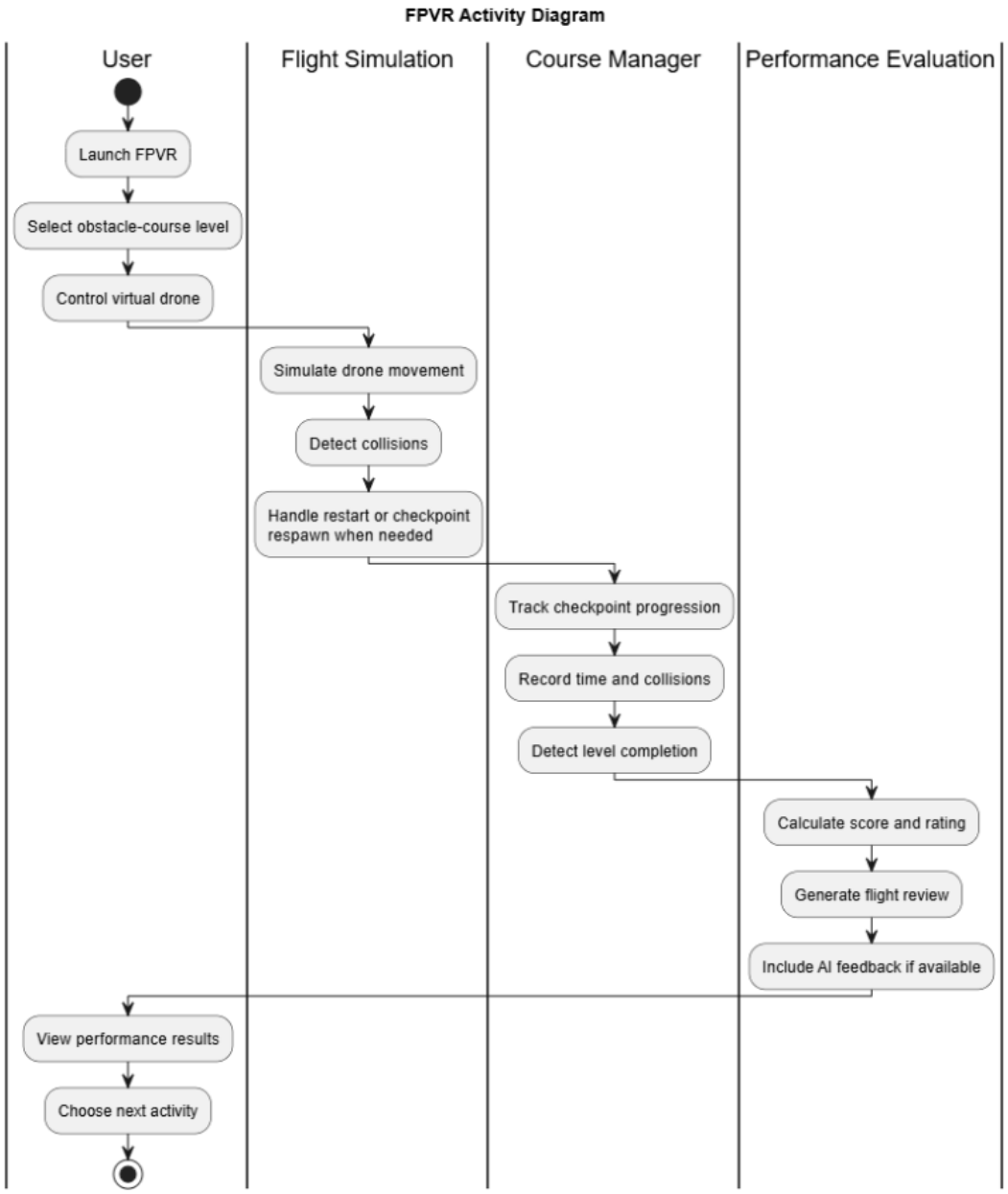


Figure 1: High-Level FPVR Training Activity Diagram

The activity diagram illustrates the primary obstacle-course training workflow. Swimlanes identify the responsible user or system component, while arrows indicate the sequence of activities.

G.6 Limitations and Exclusions

FPVR is intended to provide an introductory virtual reality (VR) training experience for first-person-view (FPV) drone operation. The following limitations and exclusions define the boundaries of the system and clarify functionality outside the project's current scope.

Table 2: FPVR Limitations and Exclusions

Area	Limitation or Exclusion
Physical Drone Operation	FPVR is limited to simulated drone operation and will not communicate with or control physical drones.
VR Hardware Requirement	FPVR requires a compatible VR headset for operation. Both Meta Quest controllers and compatible Xbox controllers will be supported, but conventional PC gameplay without a VR headset is excluded.
Pilot Certification	FPVR will not provide official drone pilot certification or replace applicable regulatory examinations, assessments, or licensing requirements.
Real-World Flight Proficiency	Performance within FPVR will not be considered proof of a user's ability to safely operate a physical drone. The transfer of skills from simulation to real-world flight is not guaranteed.
Flight Simulation Realism	FPVR aims to provide realistic flight behaviour for training purposes. However, the underlying flight model, its implementation, and the required level of accuracy remain under review. The system is not intended to reproduce every physical and environmental characteristic of real-world drone operation.
Multiplayer Functionality	FPVR is limited to single-player training. Multiplayer racing, cooperative flight, and shared online training sessions are excluded.
Dedicated FPV Transmitters	Support for dedicated FPV radio transmitters is under consideration but is not part of the confirmed core controller requirements. Compatibility and feasibility will be evaluated during development.
Historical Flight Records	FPVR will provide performance reviews after completing obstacle-course levels but will not include functionality for accessing or reviewing previously completed sessions.
AI-Assisted Feedback	AI-generated flight feedback is planned subject to feasibility evaluation. Core flight reviews, numerical scores, and performance ratings will remain available independently of this optional capability.

FPVR focuses on safe introductory flight practice, structured obstacle-course challenges, and performance evaluation within a virtual environment. The system is not intended to replace real-world flight training or serve as a comprehensive engineering-grade flight simulator. The risks associated with these limitations and their corresponding mitigation strategies are discussed in P.6.

G.7 Stakeholders and requirements sources

This section identifies the stakeholders who may influence or be affected by FPVR and the sources used to establish its initial requirements. It distinguishes confirmed project

participants from potential stakeholders whose feedback has not yet been collected.

G.7.1 Stakeholders

Table 3 identifies the primary and secondary stakeholders of FPVR and their interests in the project.

Table 3: FPVR Stakeholders

Stakeholder	**Interest in FPVR**
Beginner FPV Pilots	Primary intended users seeking accessible introductory training, opportunities to practise basic flight manoeuvres, and feedback on their performance.
Drone Enthusiasts	Potential users interested in additional flight practice, structured challenges, and evaluating the realism of the simulation.
Students and Technology Enthusiasts	Potential users interested in experiencing FPV drone operation through virtual reality. Intended use is subject to applicable headset age and safety restrictions.
McMaster Drone Team	Potential technical stakeholder that may provide feedback on simulated drone behaviour and operator training. The team has not yet been consulted.
Project Team	Responsible for developing, testing, maintaining, and documenting FPVR according to the established requirements.
Project Supervisor	Provides technical guidance and feedback on project scope, feasibility, and simulation realism.
Course Evaluators	Review the project's requirements, engineering process, documentation, and delivered functionality against the course expectations.
Parents and Guardians	Indirect stakeholders concerned with age-appropriate access and the safe use of VR by younger users.
Society and General Public	Indirect stakeholders interested in safer introductory drone training, responsible technology use, accessibility, and the management of VR-related health and safety concerns.

Society is considered an indirect stakeholder because VR-based drone training has implications for physical safety, user well-being, accessibility, and responsible drone operation. FPVR must not misrepresent simulated flight performance as proof of real-world piloting competence.

FPVR is intended for users with varying levels of experience and age, subject to the age restrictions and safety requirements of the supported VR hardware. Applicable platform age-group requirements are documented by Meta [10].

G.7.2 Requirements Sources

The initial requirements for FPVR have been developed using existing project documentation, team discussions, supervisor guidance, and technical references relevant to virtual reality and flight simulation.

The principal requirements sources are summarized in Table 4.

Table 4: FPVR Requirements Sources

Source	**Contribution to Requirements**
Problem Statement	Establishes the problem being addressed, the project's objectives, and its intended benefits.
Development Plan	Provides project scope, planned development activities, technical considerations, and project constraints.
Hazard Analysis	Identifies potential hazards and safety considerations relevant to the simulator and its users.
Supervisor Discussions and Meeting Notes	Provide technical guidance on simulation realism, drone flight behaviour, and the selection of an appropriate flight model.
Unity Documentation	Provides information about available physics simulation capabilities, including movement, forces, gravity, and collision handling [8].
Meta Quest Documentation	Provides technical information about VR device integration and controller input [9], as well as applicable age-group requirements [10].
Existing FPV Simulators	Provide a basis for comparing existing flight practice, training, and performance evaluation capabilities with the proposed functionality of FPVR.
Potential End Users	Planned source of feedback on usability, flight-control expectations, training needs, and the effectiveness of performance feedback. Direct consultation has not yet occurred.

The project supervisor has provided guidance concerning realistic flight behaviour and the selection of an existing drone model as a possible basis for the simulation. These discussions have informed the identification of technical concerns, but the final flight model and required accuracy remain under review.

No structured requirements-elicitation sessions have yet been conducted with the intended end users. The project team plans to consult potential users, including beginners and individuals familiar with drone operation, to validate assumptions about usability, training needs, and performance feedback.

Until this consultation takes place, the identified user needs and evaluation targets remain provisional. Feedback collected during development will be used to review and refine the

system requirements.

G.8 User Profiles

FPVR is intended for users with varying levels of experience in drone operation, virtual reality (VR), and video games. The system primarily targets beginners while also accommodating recreational users and experienced FPV pilots. Users must comply with the age and safety restrictions of the supported VR hardware.

The following profiles describe the anticipated experience, needs, and usage patterns of the system's main user groups.

G.8.1 Novice FPV Pilot

The primary user is an individual with little or no experience operating first-person-view (FPV) drones. This user may also be unfamiliar with VR environments and their associated controllers.

Experience: May have general familiarity with computers or video games but limited knowledge of drone flight controls, navigation, and flight behaviour.

Needs: Clear control instructions, understandable interactions, opportunities to practise without risking damage to physical equipment, and feedback on flight performance.

Expected Usage: The user reviews the basic control instructions and begins practising in free-flight mode. As confidence improves, the user may select obstacle-course levels of increasing difficulty. Progression is voluntary, as all levels are available without prerequisite completion.

G.8.2 Casual Gamer or Technology Enthusiast

This user is interested in experiencing drone flight through VR without necessarily intending to purchase or operate a physical drone.

Experience: May be familiar with video-game controllers and immersive experiences but have limited knowledge of actual FPV drone operation.

Needs: Engaging virtual environments, accessible controls, understandable objectives, and opportunities to measure improvement through repeated challenges.

Expected Usage: The user explores free-flight environments and attempts obstacle-course challenges for recreation and exploration. The user may repeat levels to improve completion

times, scores, and performance ratings.

G.8.3 Experienced FPV Pilot or Technical Evaluator

This user has prior experience operating physical FPV drones or working with related technical systems. Experienced users may participate in regular training activities or provide feedback during system evaluation.

Experience: Familiar with drone controls, flight manoeuvres, and the expected behaviour of physical drones.

Needs: Consistent control responses, meaningful obstacle-course challenges, and sufficient performance information to evaluate the simulator's behaviour.

Expected Usage: The user practises in free-flight mode, completes obstacle-course challenges, and reviews recorded performance results. The user may also compare simulated flight behaviour with prior experience and identify areas for improvement. The required accuracy of the underlying flight model remains under review.

G.8.4 General User Considerations

FPVR is intended to accommodate both seated and standing operation using a compatible VR headset. Users may interact with the simulator through Meta Quest controllers or compatible Xbox controllers.

Users may differ in physical capabilities, experience with immersive environments, and susceptibility to VR-related discomfort. The system's design will consider applicable headset safety guidance and the needs of different user groups.

The profiles presented in this section are based on the intended target audience rather than completed user interviews. As discussed in Section G.7, the project team plans to consult potential users to validate their needs, expectations, and usage assumptions.

(E) Environment

Provided description: The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate. [1]

E.1 Glossary

Provided description: Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. It introduces the terminology of the project; not just of the environment in the strict sense, but of all its parts. [1]

E.2 Components

Provided description: List of elements of the environment that may affect or be affected by the system and project. It includes other systems to which the system must be interfaced. These components may include existing systems, particularly software systems, with which the system will interact — by using their APIs (program interfaces), or by providing APIs to them, or both. These are interfaces provided to the system from the outside world. They are distinct from both: interfaces provided by the system to the outside world (S.3); and technology elements that the system's development will require (P.5). [1]

E.3 Constraints

Provided description: Obligations and limits imposed on the project and system by the environment. This chapter defines non-negotiable restrictions coming from the environment (business rules, physical laws, engineering decisions), which the development will have to take into account. [1]

E.4 Assumptions

Provided description: Properties of the environment that may be assumed, with the goal of facilitating the project and simplifying the system. It defines properties that are not imposed by the environment (like those in E.3) but assumed to hold, as an explicit decision meant to facilitate the system's construction. [1]

E.5 Effects

Provided description: Elements and properties of the environment that the system will affect. It defines effects of the system's operations on properties of the environment. Where the previous two categories (E.3, E.4) defined influences of the environment on the system, effects are influences in the reverse direction. [1]

E.6 Invariants

Provided description: Properties of the environment that the system's operation must preserve, i.e., properties of the environment that operations of the system may assume to hold when they start, and must maintain. [1]

(S) System

Provided description: The System book refines the Goals one by focusing on more detailed requirements. It describes the system itself: its components, its functionality, and the criteria by which an implementation will be judged satisfactory. [1]

S.1 Components

Provided description: Overall structure expressed by the list of major software and, if applicable, hardware parts. [1]

S.2 Functionality

This section defines the externally observable functional and non-functional requirements of FPVR. Each requirement has a unique identifier and an individual rationale. Requirements apply to the initial release unless explicitly designated as an optional enhancement. The target headset is Meta Quest 3S. Numeric thresholds and scoring parameters in this section constitute a proposed, testable baseline to be approved by the project team and verified on target hardware.

S.2.1 Drone Flight Simulation

- **FR-FLT-01:** FPVR shall display a first-person view whose position and orientation follow the simulated drone during flight.
  Rationale: The view must correspond to the vehicle being controlled to provide an FPV training experience.
- **FR-FLT-02:** FPVR shall accept independently variable throttle, pitch, roll, and yaw inputs and produce a corresponding change in simulated drone behaviour for each input.
  Rationale: These are the four primary control axes expected in manual FPV flight.
- **FR-FLT-03:** When the drone is airborne and no upward thrust is applied, its vertical velocity shall decrease under simulated gravity.
  Rationale: Gravity is necessary to distinguish the flight model from unrestricted game-style movement and can be checked without prescribing a physics engine.
- **FR-FLT-04:** FPVR shall detect drone contact with course obstacles and ground surfaces during flight.

Rationale: Contact detection is necessary to enforce obstacle challenges and initiate recovery.

- **FR-FLT-05:** Initial ground contact before the first takeoff of an attempt shall not count as a crash; contact with the ground after the drone has become airborne, or contact with a designated obstacle at any time, shall trigger a simulated crash.
  Rationale: This permits conventional takeoff while making the collision rule unambiguous. For acceptance testing, the airborne state begins when the drone is no longer in contact with its starting ground surface.

- **FR-FLT-06:** Each crash shall produce exactly one crash event, suspend normal flight input, and present the recovery behaviour applicable to the current mode.
  Rationale: A single event prevents a prolonged collision from generating duplicate penalties.

S.2.2 Training Modes and Level Management

- **FR-LVL-01:** FPVR shall provide free-flight practice without checkpoints, a completion timer, numerical scoring, or performance ratings.
  Rationale: Beginners need an unrestricted setting to learn the controls before attempting assessed challenges.

- **FR-LVL-02:** Following a crash in free-flight mode, the user shall be able to restore the drone to its free-flight starting position without restarting FPVR.
  Rationale: Quick recovery enables repeated practice without interruption.

- **FR-LVL-03:** FPVR shall provide at least three selectable obstacle-course levels, each with a distinct obstacle or checkpoint arrangement; the levels shall be identified as introductory, intermediate, and advanced.
  Rationale: Three differentiated configurations establish a verifiable minimum for varied practice without relying on subjective difficulty claims.

- **FR-LVL-04:** FPVR shall provide at least two visually distinct virtual environments, with at least one playable obstacle-course level in each.
  Rationale: Distinct environments offer variation while permitting reuse of an environment by multiple levels.

- **FR-LVL-05:** Every obstacle-course level shall be selectable immediately without prerequisites or unlock conditions.
  Rationale: Users with different skill levels should choose their own training difficulty.

- **FR-LVL-06:** The user shall be able to leave either training mode and return to the main menu without restarting the application.
  Rationale: Users must be able to stop a VR activity without terminating the software.

- **FR-LVL-07:** Starting an obstacle-course attempt shall initialize checkpoint progress to zero, both timers to zero, crash count to zero, and the attempt's score state to its initial value.
  Rationale: A fresh attempt must be independent of the previous attempt's performance.

S.2.3 Checkpoints and Crash Recovery

- **FR-CHK-01:** Each obstacle-course level shall have an ordered list of at least two checkpoints followed by a finish location.
  Rationale: Ordered segments are needed for measurable progression and checkpoint-based practice.

- **FR-CHK-02:** FPVR shall record a checkpoint as completed only when the drone passes through the next checkpoint in the prescribed order.
  Rationale: This makes completion dependent on following the intended course.

- **FR-CHK-03:** Passing through a later checkpoint before completing the next required checkpoint shall not advance progress.
  Rationale: This prevents bypassing the challenge sequence.

- **FR-CHK-04:** After a crash in an obstacle course, FPVR shall display the choices *Respawn* and *Restart Level*.
  Rationale: Users need both targeted practice and the option to begin a fresh scored attempt.

- **FR-CHK-05:** Respawn shall return the drone to the most recently completed checkpoint, or to the level start if none has been completed.
  Rationale: The player can retry the failed segment without losing earlier valid progress.

- **FR-CHK-06:** Respawn shall preserve completed checkpoints, accumulated total flight time, and the recorded crash count and score penalties.
  Rationale: Repeated failed attempts must continue to affect the overall performance result.

- **FR-CHK-07:** Restart Level shall return the drone to the level start and reset checkpoints, both timers, crash count, and score state.
  Rationale: Restarting is defined as a new attempt rather than a shortcut that retains previous penalties.

- **FR-CHK-08:** FPVR shall declare a level complete only after every checkpoint has been completed in order and the drone reaches the finish.
  Rationale: This gives a deterministic and testable definition of successful completion.

S.2.4 Timing and Performance Evaluation

- **FR-PER-01:** Each obstacle-course attempt shall have a total timer that measures accumulated active flight time from the start of the attempt to successful completion.
  Rationale: A single total time provides a comparable performance measure for a completed attempt.

- **FR-PER-02:** The total timer shall retain time spent in flight segments that end in crashes.
  Rationale: Failed attempts must not disappear from the overall completion-time result.

- **FR-PER-03:** Each obstacle-course attempt shall have a separate section timer measuring active flight time since the start of the current checkpoint segment.
  Rationale: Segment timing provides immediate feedback during repeated practice.

- **FR-PER-04:** The section timer shall reset to zero when the next required checkpoint is completed and when a checkpoint respawn occurs.
  Rationale: Each segment attempt should begin with a fresh local timer.

- **FR-PER-05:** Checkpoint respawn shall not reset the total timer; Restart Level shall reset both timers.
  Rationale: The two recovery choices intentionally have different effects on attempt history.

- **FR-PER-06:** Both timers shall pause while the crash-recovery menu is displayed and resume when active flight resumes; no menu time shall be added to either timer.
  Rationale: Decision time in a menu should not affect flight performance.

- **FR-PER-07:** FPVR shall increment the crash count once for each registered crash event in an obstacle-course attempt.
  Rationale: The count is required for meaningful penalties and a transparent review.

- **FR-PER-08:** At level completion, FPVR shall calculate a whole-number score as $\max(0, 1000 - \lfloor T \rfloor - 100C)$, where T is total active flight time in seconds and C is the number of registered crashes.
  Rationale: This proposed baseline is deterministic, easy to reproduce in tests, and penalizes both time and crashes. The constants are initial game-balance choices, not experimentally validated measures of pilot skill.

- **FR-PER-09:** FPVR shall assign the rating Gold for scores of at least 800, Silver for 600–799, Bronze for 400–599, and Needs Improvement for 0–399.
  Rationale: Fixed non-overlapping score bands make ratings unambiguous and verifiable; the bands are initial design choices rather than validated proficiency levels.

- **FR-PER-10:** The total timer shall stop when the level is successfully completed, and its final value shall be used in the score and post-flight review.
  Rationale: A fixed completion event ensures consistent reported results.

S.2.5 VR Interface and Controller Input

- **FR-UI-01:** FPVR shall present the flight view and training menus in a Meta Quest 3S headset.
  Rationale: The Quest 3S is the agreed primary evaluation device.

- **FR-UI-02:** FPVR shall support both standalone execution on Quest 3S and PCVR execution with Quest 3S connected to a compatible Windows 11 computer.
  Rationale: Both deployment paths are in the project scope; Windows 11 is the proposed reference PCVR operating system for repeatable testing.

- **FR-UI-03:** FPVR shall provide a flight-control configuration for Quest 3S motion controllers in standalone mode and an Xbox-compatible gamepad in PCVR mode, each providing throttle, pitch, roll, and yaw.
  Rationale: This defines a minimum verifiable controller-to-execution-mode matrix without promising every controller in every mode.

- **FR-UI-04:** For each supported controller configuration, FPVR shall display a control guide identifying the assigned throttle, pitch, roll, and yaw inputs before the user begins flight.
  Rationale: Explicit instructions allow users to operate the simulator without assuming previous FPV experience.

- **FR-UI-05:** FPVR shall provide menu interactions for choosing training mode, selecting a level, starting flight, recovering from crashes, and returning to the main menu.
  Rationale: Every required gameplay transition must be accessible to the user.

- **FR-UI-06:** FPVR shall permit seated and standing operation without requiring the user to physically walk to control drone movement.
  Rationale: Users should be able to operate the simulator within a stationary VR play area.

S.2.6 Player Accounts and Authentication

- **FR-ACC-01:** FPVR shall allow a user with network connectivity to register an account using a unique username and password.
  Rationale: Persistent identity is needed to associate results with a player and support global ranking.

- **FR-ACC-02:** FPVR shall reject registration when the username is already registered or the password contains fewer than 12 characters, and shall display an explanatory message.
  Rationale: Duplicate identity must be prevented, and a clear minimum password rule is easier to verify than an undefined strong-password requirement.

- **FR-ACC-03:** FPVR shall allow a registered user to sign in with valid credentials and shall reject invalid credentials without disclosing whether the username or password was incorrect.
  Rationale: Authentication protects ownership of stored results while avoiding unnecessary account enumeration.

- **FR-ACC-04:** FPVR shall provide a sign-out action that ends the current authenticated session on that installation.
  Rationale: Shared headsets or computers require a way to stop acting under another player's identity.

- **FR-ACC-05:** FPVR shall allow free-flight and obstacle-course practice without an authenticated account, but shall require authentication before submitting a result to the global leaderboard.
  Rationale: Account or network failure should not block the core training experience, while public rankings need attributable submissions.

S.2.7 Result Storage and Leaderboards

- **FR-DAT-01:** After a completed obstacle-course attempt, FPVR shall save a local result record containing the level identifier, completion timestamp, total time, crash count, score, rating, and the active local player identifier.
  Rationale: A local record supports a device leaderboard and retains results when a network connection is unavailable.

- **FR-DAT-02:** For an authenticated player with network connectivity, FPVR shall submit the completed result to the online result service and associate it with that player's account.
  Rationale: Cloud association allows results to be retrieved across installations and included in global rankings.

- **FR-DAT-03:** FPVR shall retain a locally saved result when an online submission fails and shall allow the player to retry synchronization after connectivity is restored.
  Rationale: Temporary network failure should not erase a completed attempt.

- **FR-DAT-04:** FPVR shall display a local leaderboard for each obstacle-course level using results saved on the current installation.
  Rationale: Users can compare results without relying on an external service.

- **FR-DAT-05:** FPVR shall display a global leaderboard for each obstacle-course level using results successfully accepted by the online service.
  Rationale: Online rankings enable comparison between registered players across installations.

- **FR-DAT-06:** Both leaderboards shall rank results by descending numerical score, using shorter total completion time as the first tiebreaker and earlier completion timestamp as the second tiebreaker.
  Rationale: A complete ordering rule prevents ambiguous rankings for equal scores.

- **FR-DAT-07:** Each leaderboard shall show at least the top ten available results for the selected level, including rank, player display name, score, completion time, and crash count.
  Rationale: A fixed minimum display capacity and field list make the leaderboard interface testable.

- **FR-DAT-08:** When the online leaderboard is unavailable, FPVR shall display an availability message and continue to provide the local leaderboard and training modes.
  Rationale: Online-service failure must not disable core gameplay.

- **FR-DAT-09:** A registered player shall be able to retrieve their previously synchronized completed results after signing in on another supported installation.
  Rationale: Cross-device retrieval is the user-visible benefit of cloud-backed result persistence.

- **FR-DAT-10:** FPVR shall distinguish results saved only locally from results successfully synchronized to the online service.
  Rationale: Users must not be misled into believing an offline result is already included in the global ranking.

S.2.8 Post-Flight Review

- **FR-REV-01:** FPVR shall display a post-flight review immediately after successful completion of an obstacle-course level.
  Rationale: Immediate feedback connects performance outcomes to the just-completed attempt.

- **FR-REV-02:** The review shall display the completed level, total completion time, completed checkpoints, crash count, numerical score, and rating.
  Rationale: These measures explain the result and support deliberate practice.

- **FR-REV-03:** From the review, the user shall be able to retry the same level, choose another level, or return to the main menu.
  Rationale: Users should be able to continue training without restarting the application.

- **FR-REV-04:** The standard review shall be displayed even when the network or intelligent feedback service is unavailable.
  Rationale: Core performance feedback must not depend on optional external services.

- **FR-REV-05:** FPVR shall permit registered players to view their stored completed results from prior sessions, showing at least the fields specified in FR-DAT-01.
  Rationale: Account-linked result persistence is useful only if users can retrieve and inspect their earlier performance; this supersedes the earlier exclusion of historical results.

S.2.9 Intelligent Feedback

- **FR-AI-01:** FPVR shall generate a natural-language observation based on the completed attempt's available time, checkpoint, and crash data.
  Rationale: Feedback must be grounded in measurable attempt data rather than invented flight events.

- **FR-AI-02:** FPVR shall display it automatically in the post-flight review without requiring an additional user request.
  Rationale: Automatic presentation makes the enhancement accessible during the normal review workflow.

- **FR-AI-03:** Failure or unavailability of intelligent feedback shall not prevent display of the standard post-flight review.
  Rationale: An add-on feature must not become a single point of failure for required feedback.

S.2.10 Non-Functional Requirements

- **NFR-INST-01 (Installability):** FPVR shall install and launch as a standalone application on Meta Quest 3S without a connected PC.
  Rationale: Standalone operation is a committed delivery mode, not merely an editor demonstration.

- **NFR-INST-02 (Installability):** FPVR shall launch in PCVR mode on a Windows 11 computer connected to a Quest 3S using a supported PCVR connection method.
  Rationale: A specified reference operating system makes the second deployment mode independently testable.

- **NFR-COMP-01 (Compatibility):** Quest 3S motion controllers shall support the four flight axes in standalone mode, and an Xbox-compatible gamepad shall support the four flight axes in PCVR mode.

Rationale: This minimum matrix is testable without promising unsupported controller combinations.

- **NFR-PERF-01 (Rendering):** On Quest 3S in standalone mode, FPVR shall maintain an average of at least 72 rendered frames per second over a continuous 120-second normal-flight test in each of the two required environments.
  Rationale: The 72-FPS threshold is an initial headset-oriented performance target; averaging over a fixed interval makes the result reproducible, although frame-time spikes should also be inspected during V&V.

- **NFR-PERF-02 (Input Response):** In the reference test environment, the elapsed time from a received flight-control input event to the first corresponding simulated state update shall not exceed 100 ms in at least 95% of 100 measured input events.
  Rationale: This proposed software-response threshold makes responsiveness measurable without conflating it with headset display latency.

- **NFR-REL-01 (Reliability):** FPVR shall complete 20 consecutive crash-and-respawn cycles in each required obstacle-course environment without application termination or loss of checkpoint consistency.
  Rationale: A repeatable stress test checks the recovery workflow that users will exercise frequently.

- **NFR-USAB-01 (Usability):** The control guide shall be accessible from the main menu and before the first flight of a newly launched session.
  Rationale: Beginners need discoverable instructions without requiring prior experience.

- **NFR-SAFE-01 (VR Safety):** FPVR shall allow the user to pause or leave an active flight activity using an in-application menu without removing the headset.
  Rationale: Users must be able to stop an uncomfortable experience promptly.

- **NFR-SAFE-02 (Stationary Use):** All core flight, menu, and recovery actions shall be operable while seated or standing in one physical location.
  Rationale: Stationary operation reduces the need for movement within the user's physical surroundings.

- **NFR-SEC-01 (Credential Protection):** FPVR shall transmit account credentials only over encrypted connections and shall not store account passwords in plaintext on the device or online service.
  Rationale: Online account functionality introduces credential exposure risks absent from the original offline-only simulator.

- **NFR-SEC-02 (Access Control):** An unauthenticated user shall not be able to retrieve another player's non-public account data or submit a leaderboard result under another player's account identifier.
  Rationale: Public rankings must not compromise account privacy or permit identity spoofing.

- **NFR-DAT-01 (Persistence):** A result confirmed as synchronized shall remain retrievable after the player signs out, restarts FPVR, and signs in again.
  Rationale: Persistence across sessions is the defining acceptance condition for cloud-backed storage.

- **NFR-OFF-01 (Offline Availability):** Free-flight, obstacle-course play, crash recovery, local result saving, and standard post-flight review shall remain usable when the online service is unavailable.
  Rationale: Core training should remain available despite network failures; account registration, global ranking, and synchronization may require connectivity.

Baseline interpretation. The requirements above define the proposed initial acceptance baseline. The numerical score, rating bands, frame-rate target, input-response threshold, and reference PCVR platform are initial project specifications rather than empirically validated claims. Any later change shall be recorded in revision history and propagated to verification tests, traceability links, and the affected scope sections.

Requirement priorities will be established in Section S.5, while verification and acceptance criteria will be defined in Section S.6.

S.3 Interfaces

Provided description: How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs). It specifies how that functionality will be made available to the rest of the world, including people (users) and other systems. These are interfaces provided by the system to the outside; the other way around, interfaces from other systems, which the system may use, are specified in E.2. [1]

S.4 Detailed usage scenarios

Provided description: Examples of interaction between the environment (or human users) and the system, expressed as user stories. Such scenarios are not by themselves a substitute for precise descriptions of functionality (S.3), but provide an important complement by specifying cases that these behavior descriptions must support; they also serve as a basis for developing test cases. The scenarios most relevant for stakeholders are given in chapter G.5 in the Goals book, at a general level, as use cases; in contrast, S.4 can refer to system components and functionality (from other chapters of the System book) as well as special and erroneous cases, and introduce more specific scenarios. [1]

S.5 Prioritization

Provided description: Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality. It is useful in particular if during the course of the project various pressures force the team to drop certain functions. [1]

S.6 Verification and acceptance criteria

Provided description: Specification of the conditions under which an implementation will be deemed satisfactory. Here, "verification" as shorthand for what is more explicitly called "Verification & Validation" (V&V), covering several levels of testing — module testing, integration testing, system testing, user acceptance testing — as well as other techniques such as static analysis and, when applicable, program proving. [1]

(P) Project

Provided description: The Project book describes not the system but the process that produces it: roles, schedule, tasks and deliverables, required technology, and the risks that could derail it. [1]

P.1 Roles and personnel

Provided description: Main responsibilities in the project; required project staff and their needed qualifications. It defines the roles (as a human responsibility) involved in the project. [1]

P.2 Imposed technical choices

Provided description: Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters. Not all technical choices in projects derive from a pure technical analysis; some result from company policies. While some project members may dislike non-strictly-technical decisions, they are a fact of project life and must be documented, in particular for the benefit of one of the quality factors for requirements: "requirements must be justified". [1]

P.3 Schedule and milestones

Provided description: List of tasks to be carried out and their scheduling. It defines the project's key dates. [1]

P.4 Tasks and deliverables

Provided description: This is the core of the Project book. It details the individual tasks listed under P.3 and their expected outcomes. It define the project's main activities and the results they must produce, associated with the milestone dates defined in P.3. [1]

P.5 Required technology elements

Provided description: External systems, hardware and software, expected to be necessary for building the system. It lists external technology elements, such as program libraries and hardware

devices, that the project is expected to require. Although the actual use of such products belongs to design and implementation rather than requirements, it is part of the requirements task to identify elements whose availability is critical to the success of the project — an important element of risk analysis (P.6). [1]

P.6 Risk and mitigation analysis

Provided description: Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise. It is essential to be on the lookout for events that could derail the project, and devise mitigation strategies. It can include a SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) for the project. [1]

P.7 Requirements process and report

Provided description: Initially, description of what the requirements process will be; later, report on its steps. It starts out as a plan for conducting the requirements elicitation process, but is meant to be updated as part of that process so that it includes the key lessons of elicitation. [1]

References

[1] Bertrand Meyer. *Handbook of Requirements and Business Analysis*. Springer, 2022.

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[4] VelociDrone. *VelociDrone FPV Simulator: Features*. Available: https://www.velocidrone.com/features. Accessed: October 18, 2026.

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[6] Meta Platforms, Inc. *DVR Simulator*. Meta Quest Store. Available: https://www.meta.com/experiences/dvr-simulator/3744737298949811/. Accessed: October 18, 2026.

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[8] Unity Technologies. *Rigidbody Physics*. Unity 6.0 User Manual. Available: https://docs.unity3d.com/6000.0/Documentation/Manual/rigidbody-physics-section.html. Accessed: October 18, 2026.

[9] Meta Platforms, Inc. *Controller Input and Tracking Overview*. Meta Horizon OS Developer Documentation. Updated: February 13, 2026. Available: https://developers.meta.com/vr/documentation/unity/unity-ovrinput/. Accessed: October 18, 2026.

[10] Meta Platforms, Inc. *Age Group Self-Certification and Youth Requirements*. Meta Horizon OS Developer Documentation. Updated: April 2, 2026. Available: https://developers.meta.com/vr/resources/age-groups/. Accessed: October 18, 2026.

Appendix — Reflection

[Not required for CAS 741 —SS]

The information in this section will be used to evaluate the team members on the

graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?